Maximum Entropy Record Linkage with Continuous Comparisons

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Abstract

In this paper, we generalize the unsupervised maximum entropy classification (MEC) algorithm for record linkage, originally proposed by Lee et al. (2022, Surv. Methodol.), to allow for the use of continuous comparison functions. The initial MEC approach relies on binary comparisons, thereby overlooking small discrepancies between key variable values. We incorporate continuous comparison functions by developing two approaches: parametric and nonparametric. The former is based on the hurdle Gamma distribution (cf. Vo et al., 2023, Comput. Stat. Data Anal.), whereas the latter uses the Kullback-Leibler Importance Estimation Procedure (cf. Sugiyama et al., 2008, Ann. Inst. Stat. Math.). Next, we integrate our method with probabilistic blocking to ensure computational scalability. We conduct two simulation studies, which indicate that the proposed continuous approaches perform well and offer overall improvements over the binary algorithm in most cases. Finally, we evaluate the developed method by linking two administrative registers of foreign-born nationals in Poland. The proposed approaches are implemented in the `automatedRecLin` R package, available on CRAN.

Keywords: record linkage, entity resolution, maximum entropy classification, continuous comparison functions, density ratio

1 Introduction

Many modern statistical analyses require combining data from multiple sources. The process of linking such data on the same entity is called *entity resolution* (ER; Papadakis et al., 2021) or *probabilistic record linkage* (Dunn, 1946; Fellegi and Sunter, 1969). The core assumption is that it is possible to identify the records associated with the same entity across sources. Ideally, data sets contain unique keys (identifiers), but this is rarely the case, and one has

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to use *key* variables such as first name, last name, or date of birth. In this paper, we focus on record linkage (RL) between two sources.

Record linkage has been applied in many research fields, such as official statistics, social sciences, or medicine (cf. Binette and Steorts, 2022). The first automated, computer-supported method was introduced by Newcombe et al. (1959), and Fellegi and Sunter (1969) formalized the problem by proposing an approach based on the likelihood ratio test. Their idea requires estimating model parameters and setting thresholds for establishing decision rules. The former issue was addressed by Winkler (1988) and Jaro (1989), who proposed an expectation-maximization (EM) algorithm. However, a disadvantage of the approach by Fellegi and Sunter (1969) is that some pairs need to be verified and classified by clerical review. Additionally, it assumes independence between measures of agreement of different record pairs, but in real situations, this assumption is rarely fulfilled; see Binette and Steorts (2022) for further discussion.

Lee et al. (2022) note that RL is a classification problem, where each record pair is classified either as a *match* or a *nonmatch*. They propose adapting the maximum entropy classification (MEC) to record linkage. MEC was also previously used by Nigam et al. (1999) for text classification, while Nguyen et al. (2010) developed the theory of the maximum entropy method for density ratio estimation. The key difficulty of applying MEC in the context of record linkage is that records from both data sets belong to many record pairs. Lee et al. (2022) consider supervised and unsupervised algorithms, which rely on binary comparisons. Their approach can be viewed as an extension of the seminal work of Fellegi and Sunter (1969) and it addresses two fundamental problems: the need to evaluate all record pairs simultaneously rather than independently, and the dependence between measures of agreement of different record pairs.

In this paper, we propose an extension of the MEC record linkage algorithm that incorporates continuous comparisons between key variables. For character variables, one may use the Jaro-Winkler similarity measure (Winkler, 1990), while for numerical variables, the Euclidean distance can be considered. Our main contribution is based on the hurdle Gamma distribution (cf. Choi and Wette, 1969; Vo et al., 2023), which was previously adopted for the standard record linkage method introduced by Fellegi and Sunter (1969). Here, we propose two approaches: a *continuous parametric* algorithm and a *continuous nonparametric* algorithm. The former is based on the aforementioned hurdle Gamma distribution, whereas the latter uses the Kullback-Leibler Importance Estimation Procedure (KLIEP; see Sugiyama et al., 2008). Finally, to address the scalability problem, we propose integrating MEC for record linkage with a probabilistic blocking approach based on approximate nearest neighbor (ANN) algorithms and graphs to create relatively small blocks that allow for the computationally efficient estimation of the parameters of the MEC algorithm.

We conduct two simulation studies and present an application. The first simulation demonstrates MEC's ability to perform high-quality record linkage. The proposed parametric method outperforms the original approach introduced by Lee et al. (2022) in terms of error rates, as well as in estimating the number of pairs that should be linked. The continuous nonparametric algorithm yields better results than the binary method, except in terms of linkage precision. In the second simulation, we show that choosing between binary and continuous comparisons in MEC with blocking entails a tradeoff: binary comparisons yield fewer missing matches, whereas continuous comparisons result in fewer false

links. Additionally, we evaluate MEC-based methods against the classical model developed by Fellegi and Sunter (1969). Finally, we present a case study using administrative data on foreign-born nationals in Poland, linking records from the register of the Office for Foreigners (UDSC) about legalization of stay with the Social Insurance Institution (ZUS) register. This linkage allows us to provide more information on migrants in the Polish labor market.

This paper is organized as follows. In Section 2, we introduce the notation and the MEC approach to record linkage. In Section 3, we describe the proposed method. In Section 4, we discuss the simulation studies and their results. In Section 5, we apply the developed approach to data on foreign-born nationals in Poland for 2023. Finally, in Section 6, we conclude the paper. Details of the algorithms, the simulation studies, and the case study are presented in the appendices.

2 Basic Setup

We focus on unsupervised maximum entropy classification for record linkage. Due to the lack of training data, unsupervised approaches are the most common in practice. However, both the binary algorithm by Lee et al. (2022) and our methods incorporating continuous comparisons can be adjusted to the supervised setting. For details on the supervised MEC for record linkage, see Lee et al. (2022).

2.1 Notation and Conditional Independence Among Agreement Indicators

In this section, we follow the notation used by Lee et al. (2022). Consider two data sets, A and B. We assume that they share many entities but contain no internal duplicates. Our goal is to link A to B. Let the bipartite comparison space $\Omega = A \times B = M \cup U$ consist of *matches* M and *nonmatches* U between the records in A and B. Let $n = |\Omega|$ be the number of all record pairs, and $n_M = |M|$ be the number of matches. The matching indicator is denoted by g_{ab}, that is, $g_{ab} = \mathbb{I}((a, b) \in M)$. For any pair of records $(a, b) \in \Omega$, let $\boldsymbol{\gamma}_{ab} = (\gamma_{ab}^1, \gamma_{ab}^2, \ldots, \gamma_{ab}^K)'$ be the comparison vector based on a set of K key variables. The key variables must be present in both data sets but may be affected by errors. For example, in applications related to human populations, researchers may use name, sex, or date of birth.

In all algorithms considered, we assume conditional independence (given g_{ab}) among the components of $\boldsymbol{\gamma}$. However, in practice, this assumption might not hold. For instance, in an administrative register, a typo in the first name of a foreign-born person might increase the probability of observing a typo in their last name. Nonetheless, even when this assumption is violated, the results might not be severely affected; for example, see Winkler (1994).

Let $m(\boldsymbol{\gamma})$ and $u(\boldsymbol{\gamma})$ be the conditional probability or density functions of $\boldsymbol{\gamma} = \boldsymbol{\gamma}_{ab}$ given $g_{ab} = 1$ and $g_{ab} = 0$, respectively. Additionally, let $\mathcal{S}(M)$ and $\mathcal{S}(U)$ be the supports of $\boldsymbol{\gamma}_{ab}$ over M and U, respectively. Assuming $\mathcal{S}(M) \subseteq \mathcal{S}(U)$, we can define the probability or density ratio as

$$r(\boldsymbol{\gamma}) = \frac{m(\boldsymbol{\gamma})}{u(\boldsymbol{\gamma})}.$$

The MEC record linkage algorithm proposed by Lee et al. (2022) aims to minimize the following objective function with respect to r:

$$Q = \sum_{(a,b)\in M} \frac{u(\boldsymbol{\gamma}_{ab})}{n_M(\boldsymbol{\gamma}_{ab})} r(\boldsymbol{\gamma}_{ab}) - \frac{1}{n_M} \sum_{(a,b)\in M} \log r(\boldsymbol{\gamma}_{ab}),$$

where $n_M(\boldsymbol{\gamma}_{ab}) = \sum_{(i,j)\in M} \mathbb{I}(\boldsymbol{\gamma}_{ij} = \boldsymbol{\gamma}_{ab})$.

2.2 Classification Sets for Record Linkage and Linkage Errors

Provided there are no duplicates in either A or B, if a record from either data set belongs to more than one pair from those selected as matches, it automatically results in at least one false link. Therefore, Lee et al. (2022) define a *classification set* for record linkage, denoted by $\widehat{M}$, as a subset of Ω such that any record from A or B appears in at most one record pair within $\widehat{M}$. Its entropy is given by

$$D_{\widehat{M}} = \frac{1}{|\widehat{M}|} \sum_{(a,b)\in\widehat{M}} \log r(\boldsymbol{\gamma}_{ab}).$$

Lee et al. (2022) propose the following greedy method for constructing the classification set.

Definition 1 (MEC set) *A MEC set of a given size $n^* = |\widehat{M}|$ is the first classification set of size n^*, obtained by deduplication in descending order of $r(\boldsymbol{\gamma}_{ab})$ over Ω.*

In this framework, it is possible that $(a, b') \notin \widehat{M}$ even if $r(\boldsymbol{\gamma}_{ab'}) > r(\boldsymbol{\gamma}_{a'b'})$ for some $(a', b') \in \widehat{M}$. This occurs if there exists $(a, b) \in \widehat{M}$ with $r(\boldsymbol{\gamma}_{ab}) > r(\boldsymbol{\gamma}_{ab'})$. Additionally, let the maximal MEC set M_1 be the subset of Ω consisting of record pairs with perfect and unique agreement on all key variables.

In record linkage, we consider two types of errors: a *false link* if a record pair is incorrectly linked, and a *missing match* if a pair of records referring to the same entity is missed by a linkage procedure. The *false linkage rate* (FLR) is given by

$$\text{FLR} = \frac{1}{|\widehat{M}|} \sum_{(a,b)\in\widehat{M}} (1 - g_{ab}), \tag{1}$$

while the *missing match rate* (MMR) is defined as

$$\text{MMR} = 1 - \frac{1}{n_M} \sum_{(a,b)\in\widehat{M}} g_{ab}. \tag{2}$$

2.3 Binary Approach

For the comparison vectors, Lee et al. (2022) propose using K binary agreement indicators, defined as $\gamma_{ab}^k = 1$ if records $a \in A$ and $b \in B$ agree on the k-th variable, and $\gamma_{ab}^k = 0$ otherwise. For $k = 1, 2, \ldots, K$, let

$$\theta_k = \mathbb{P}(\gamma_{ab}^k = 1 \mid g_{ab} = 1) \quad \text{and} \quad \xi_k = \mathbb{P}(\gamma_{ab}^k = 1 \mid g_{ab} = 0)$$

be the conditional probabilities of agreement on the k-th variable. The parameter vectors are denoted by

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K) \quad \text{and} \quad \boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_K).$$

Assuming conditional independence among the components of the comparison vector, we obtain

$$m(\boldsymbol{\gamma}; \boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\gamma^k} (1 - \theta_k)^{1-\gamma^k} \quad \text{and} \quad u(\boldsymbol{\gamma}; \boldsymbol{\xi}) = \prod_{k=1}^{K} \xi_k^{\gamma^k} (1 - \xi_k)^{1-\gamma^k}.$$

The unsupervised MEC algorithm proposed by Lee et al. (2022) assumes a common support $\mathcal{S}(M) = \mathcal{S}(U)$ and is iterative. Initially, since $\Omega = M \cup U$ is highly imbalanced (the prevalence of $g_{ab} = 1$ is very close to 0), it estimates $\boldsymbol{\xi}$ using all pairs from Ω. Subsequently, it iteratively estimates $\boldsymbol{\theta}$ (based on the current set of matches), predicts the number of matches, and creates a new classification set (following the procedure described in Definition 1). To estimate n_M, the formula

$$n_M = \sum_{(a,b) \in \Omega} \hat{g}(\boldsymbol{\gamma}_{ab}) \tag{3}$$

is applied, where

$$\hat{g}(\boldsymbol{\gamma}_{ab}) = \min\left\{ \frac{|M| r(\boldsymbol{\gamma}_{ab})}{|M|(r(\boldsymbol{\gamma}_{ab}) - 1) + n}, 1 \right\},$$

and $|M|$, $r(\boldsymbol{\gamma}_{ab})$ are based on estimates from the previous iteration.

3 The Proposed Approach: Incorporating Continuous Comparisons

A limitation of the algorithm outlined in Section 2.3 is the loss of information about records that have similar (but not equal) values of some key variables. Suppose that we link records using the variables *first name* and *last name*. Consider the records (*Thomas*, *Smith*) and (*James*, *Wilson*) from A and (*Tomas*, *Smyth*) from B. Since they do not share the same key variable values, comparing them using binary indicators results in two comparison vectors $(0, 0)'$. On the other hand, intuitively, the records (*Thomas*, *Smith*) and (*Tomas*, *Smyth*) might refer to the same person, while the records (*James*, *Wilson*) and (*Tomas*, *Smyth*) are not likely to be a match.

We propose the use of continuous comparison functions that are semi-metrics along with binary measures. This allows us to create vectors of mixed comparisons where the binary MEC is a special case. For character variables, one may use string similarity scores, such as the Jaro-Winkler similarity (Winkler, 1990). It takes values between 0 and 1, where higher values indicate greater similarity. Hence, we can construct the semi-metric 1 − Jaro-Winkler similarity, which will later be applied in our simulation and case studies. For numerical variables, one can consider the Euclidean distance. See Vo et al. (2023) for an application to dates.

Our contribution consists of two generalizations of the MEC approach. In Section 3.1, we introduce the parametric algorithm, whereas in Section 3.2, we describe the nonparametric one. Additionally, Appendix A provides a discussion on the convergence of the unsupervised MEC algorithm proposed in the original paper.

3.1 Continuous Parametric Algorithm

Vo et al. (2023) consider incorporating continuous comparisons into the classical RL model introduced by Fellegi and Sunter (1969). They propose the use of the hurdle Gamma distribution, which is defined as follows.

Definition 2 (Hurdle Gamma distribution) *The density function of the hurdle Gamma distribution is characterized by three parameters, $p_0 \in (0,1)$ and $\alpha, \beta > 0$, as follows:*

$$f(x; p_0, \alpha, \beta) = p_0^{\mathbb{I}(x=0)}[(1-p_0)v(x;\alpha,\beta)]^{\mathbb{I}(x>0)},$$

where $x \geq 0$ and

$$v(x;\alpha,\beta) = \frac{\beta^\alpha x^{\alpha-1}\exp(-\beta x)}{\Gamma(\alpha)}$$

is the density function of the Gamma distribution.

Incorporating this distribution into the MEC procedure requires estimating its parameters. To estimate p_0, we can use the proportion of zeros in the sample. Maximum likelihood estimation of the Gamma distribution parameters is described by Choi and Wette (1969).

Assume that the components of $\boldsymbol{\gamma}$ are conditionally independent (both in M and U). Let $\gamma^k \mid M$ and $\gamma^k \mid U$ follow hurdle Gamma distributions with parameters $p_{0M}^k, \alpha_M^k, \beta_M^k$ and $p_{0U}^k, \alpha_U^k, \beta_U^k$, respectively. Let $\boldsymbol{\theta}$ and $\boldsymbol{\xi}$ denote the parameter vectors associated with M and U, respectively. In this setting, we have

$$m(\boldsymbol{\gamma};\boldsymbol{\theta}) = \prod_{k=1}^{K} f(\gamma^k; p_{0M}^k, \alpha_M^k, \beta_M^k) \quad \text{and} \quad u(\boldsymbol{\gamma};\boldsymbol{\xi}) = \prod_{k=1}^{K} f(\gamma^k; p_{0U}^k, \alpha_U^k, \beta_U^k). \tag{4}$$

Assume a common support $\mathcal{S}(M) = \mathcal{S}(U)$ and let M_1 be the maximal MEC set. Algorithm 1 provides an outline of the continuous parametric method.

Algorithm 1 Continuous parametric algorithm

1: Initialize $\boldsymbol{\theta}^{(0)}$ and set $M^{(0)} = M_1$.
2: Based on all pairs from Ω, estimate p_{0U}^k as the proportion of zeros in the sample, and estimate α_U^k, β_U^k based on the formulas from Choi and Wette (1969).
3: Given $\hat{\boldsymbol{\xi}}$, calculate $n_M^{(0)}$ using (3).
4: Using the current density ratio values and following the procedure described in Definition 1, find the MEC set $M^{(1)}$ such that $|M^{(1)}| = n_M^{(0)}$.
5: **Repeat** for $t = 1, 2, \ldots$
6: Update the estimates of p_{0M}^k, α_M^k, and β_M^k based on $M^{(t)}$.
7: Calculate $n_M^{(t)}$.
8: Find the MEC set $M^{(t+1)}$.
9: **Until** $|n_M^{(t)} - n_M^{(t-1)}| < \delta$ or $\|\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^{(t-1)}\| < \varepsilon$, where δ and ε are pre-defined thresholds (small positive values).

A detailed description of the algorithm is presented in Appendix B.

3.2 Continuous Nonparametric Algorithm

Although the continuous parametric approach performs well in our subsequent simulation and case studies, it requires the comparison variables to follow the hurdle Gamma distribution. In practice, this might not be the case. Therefore, one can also consider a nonparametric method. Instead of requiring $\boldsymbol{\gamma}$ to follow a particular distribution, we can assume that each comparison variable γ^k has a hurdle distribution. That is, it has a positive probability of taking the value 0 and follows a continuous distribution otherwise. Given this, r can be written as

$$r(\boldsymbol{\gamma}) = \prod_{k=1}^{K} \left\{ \left(\frac{p_{0M}^k}{p_{0U}^k} \right)^{\mathbb{I}(\gamma^k=0)} \left(\frac{1-p_{0M}^k}{1-p_{0U}^k} r_+^k(\gamma^k) \right)^{\mathbb{I}(\gamma^k>0)} \right\}, \tag{5}$$

where p_{0M}^k and p_{0U}^k are unknown parameters, and $r_+^k(\cdot)$ is an unknown function.

To modify the algorithm presented in Section 3.1, we need to use a method that allows for the direct estimation of the density ratio between two continuous distributions. For this purpose, we employ the Kullback-Leibler Importance Estimation Procedure (KLIEP; Sugiyama et al., 2008). KLIEP models the ratio as a weighted sum of basis functions and formulates an objective function (based on the Kullback-Leibler divergence) to be maximized with respect to the weights. This formulation leads to a convex optimization problem, which is solved by iteratively performing gradient ascent subject to a constraint. The KLIEP method is implemented in the `densityratio` R package (Volker et al., 2024). We use this approach to update $r_+^k(\gamma^k)$, while p_{0M}^k and p_{0U}^k are estimated using the proportion of zeros in the respective sample.

The basic idea behind the continuous nonparametric algorithm is similar to that of Algorithm 1, and thus the entire procedure is presented in Appendix C.

3.3 MEC with Blocking

We combine MEC with blocking methods to reduce the number of considered comparisons. In particular, we use ANN algorithms and graphs to create numerous small blocks that can be used in the actual record linkage process. For details on the blocking approach, we refer the reader to Beręsewicz and Struzik (2026).

3.3.1 Blocking Notation

Suppose that a blocking procedure returns H blocks $\Omega_1, \Omega_2, \ldots, \Omega_H$. We have $\Omega_h = A_h \times B_h$, where A_h and B_h contain records from A and B, respectively, which are assigned to the h-th block. Each record from either data set can belong to at most one block. The goal of blocking is to exclude from the comparison space pairs that definitely should not be matched. However, if a true match is not included in any block, it will constitute a missing match.

After blocking, the new comparison space is

$$\Omega_B = \bigcup_{h=1}^{H} \Omega_h.$$

It contains $N_B = \sum_{h=1}^{H} |A_h||B_h|$ record pairs. We have $\Omega_B = M_B \cup U_B$, where $M_B = M \cap \Omega_B$ and $U_B = \Omega_B \setminus M_B$. Ideally, blocking preserves all true matches, so $M_B = M$. However, in

reality, this might not always be the case. Provided there are no duplicates in either A or B, the upper bound on the number of matches in the blocks is $\nu = \sum_{h=1}^{H} \min\{|A_h|, |B_h|\}$.

After constructing Ω_B, we create comparison vectors for all pairs in Ω_B. These can be based on combinations of binary and continuous comparisons. For notational simplicity, in this section, both $m(\boldsymbol{\gamma})$ and $u(\boldsymbol{\gamma})$ are conditional on $(a, b) \in \Omega_B$. When integrating MEC with blocking, we assume that $\gamma^k \mid M_B$ and $\gamma^k \mid U_B$ follow Bernoulli or hurdle Gamma distributions. Let $\boldsymbol{\theta}$ and $\boldsymbol{\xi}$ denote the parameter vectors associated with M_B and U_B, respectively.

3.3.2 Problems with a Direct Application of MEC

The size of the blocks and, consequently, the number of pairs in the newly created comparison space depend on the applied blocking procedure. Deterministic methods, such as blocking records based on exact agreement on variables like sex, may still result in large blocks. However, we focus on probabilistic blocking that yields numerous small blocks.

As mentioned in Section 2.3, in record linkage, the full comparison space is highly imbalanced and dominated by nonmatches. It follows that if we use all pairs from Ω, our goal is to identify a very small set of matches among a large number of nonmatches. By contrast, if we consider Ω_B instead of Ω, the situation changes. The comparison space becomes enriched with matches, and the goal is to *filter out* the remaining nonmatches. Note that these nonmatches typically consist of highly similar record pairs, as they were assigned to the same block.

3.3.3 Blocked MEC for Record Linkage

To address the goal of filtering out nonmatches, we propose to *invert* the MEC approach. That is, instead of r, we consider

$$s(\boldsymbol{\gamma}) = \frac{u(\boldsymbol{\gamma})}{m(\boldsymbol{\gamma})}.$$

Therefore, pairs with higher values of s are more nonmatch-like.

The first part of the algorithm concerns the initialization of $M_B^{(0)}$. Since we expect Ω_B to be match-enriched, we can initialize the set of matches to be of size ν. Therefore, for each $(a, b) \in \Omega_B$ and $k = 1, 2, \ldots, K$, we define d_{ab}^k as

$$d_{ab}^k = \begin{cases} 1 - \gamma_{ab}^k, & \text{if } \gamma_{ab}^k \text{ is a result of a binary comparison,} \\ \gamma_{ab}^k, & \text{if } \gamma_{ab}^k \text{ is a result of a continuous comparison.} \end{cases}$$

Hence, for each key variable, $d_{ab}^k = 0$ if and only if the records $a \in A$ and $b \in B$ share the same value of this variable. Next, we define d_{ab} as

$$d_{ab} = \sqrt{\sum_{k=1}^{K} (d_{ab}^k)^2}$$

and use this quantity to construct the initial set of matches. Namely, we compute $M_B^{(0)}$ as the first set of size ν obtained by deduplication in ascending order of d_{ab} over Ω_B. Based on $M_B^{(0)}$, we can obtain an initial estimate of $\boldsymbol{\theta}$. Note that one could also consider other

initialization procedures. For example, if continuous comparisons yield values well above 1, one may scale d_{ab}^k to be between 0 and 1.

Since Ω_B may no longer be dominated by nonmatches, we cannot repeat the standard MEC approach (used on the full comparison space) of estimating $\boldsymbol{\xi}$ once at the beginning of the algorithm. Furthermore, note that the pairs in U_B have passed through blocking, which means that they are not *definitive* nonmatches. That is, the comparison vectors of some of these pairs are most likely not very different from those representing true matches with many discrepancies or typos. Hence, in some cases, even estimating $\boldsymbol{\xi}$ based on all pairs from $U_B^{(t)}$ may result in $u(\boldsymbol{\gamma})$ being too similar to $m(\boldsymbol{\gamma})$.

After constructing $M_B^{(0)}$, let $U_B^{(0)} = \Omega_B \setminus M_B^{(0)}$. Since at this initialization point, there is no information on the probability that a given pair is a match or nonmatch, we use $U_B^{(0)}$ to obtain an initial estimate of $\boldsymbol{\xi}$. However, for subsequent iterations, we propose defining a value $\rho \in [0, 1)$, which will control the fraction of pairs *excluded* from $U_B^{(t)}$ when estimating $\boldsymbol{\xi}$. That is, the t-th iteration ($t \geq 1$) of the MEC algorithm is as follows.

1. Update $\boldsymbol{\theta}$ based on $M_B^{(t)}$.
2. Take a subset $U_B^{\prime(t)} \subseteq U_B^{(t)}$ consisting of $\lfloor \rho |U_B^{(t)}| \rfloor$ pairs with the highest values of $\hat{g}_{ab}^{(t-1)}$ and use $U_B^{(t)} \setminus U_B^{\prime(t)}$ to estimate $\boldsymbol{\xi}$.
3. Calculate $n_{M_B}^{(t)}$ as an estimate of $|M_B|$.
4. Find $M_B^{(t+1)}$ of size $n_{M_B}^{(t)}$ by deduplication in ascending order of $s(\boldsymbol{\gamma}_{ab})$ over Ω_B.

The algorithm combining blocking with MEC is detailed in Appendix D. Note that if $\rho = 0$, in each iteration, we use all pairs from $U_B^{(t)}$ to estimate $\boldsymbol{\xi}$. This should be treated as a default approach. However, setting $\rho > 0$ may, in some cases, substantially improve error rates. We return to this issue in Section 4.2 while discussing one of our simulations.

3.4 MEC Implementation

We implemented the MEC-based record linkage algorithms in the `automatedRecLin` R package (Struzik and Beręsewicz, 2026; R Core Team, 2026). The software is available on CRAN under the GNU General Public License, version 3. The package allows users to perform maximum entropy classification for record linkage in both supervised and unsupervised settings. We encourage readers to refer to the package documentation for details on its functionalities. The source code is available on GitHub at `https://github.com/ncn-foreigners/automatedRecLin`.

3.4.1 Standard MEC Implementation

The core function of the package is `mec()`, which operates in the unsupervised setting and performs record linkage on all possible record pairs between two data sets. For each key variable, the user can select a comparison function and an estimation method using the `comparators` and `methods` arguments. Such an approach is useful in real applications, where continuous comparators can naturally be applied to variables such as *first name*

or *last name*, whereas for *sex* or other categorical variables, binary comparisons may be preferred. Additionally, the function implements a procedure that allows the user to have greater control over the FLR or MMR by selecting their target values.

The package also enables the evaluation of the methods using labeled data sets via the `true_matches` argument. When provided with a table containing indices of the true matches, the function automatically computes the FLR and MMR.

3.4.2 Blocked MEC Implementation

We implemented the algorithm outlined in Section 3.3.3 in the `automatedRecLin` package as the `mec_blocking()` function. Its syntax is similar to that of the `mec()` function. Note that the `blocking` package performs an ANN search based on data sets represented either as single character columns or as numeric matrices (such as embeddings).

To use the first option, the values of the blocking variables can be concatenated. This is achieved by providing a vector of key variable names to the `blocking_variables` argument. Users can also specify a custom separator via the `blocking_sep` parameter (the default value is a space). Alternatively, numeric matrices can be passed directly to the function using the `blocking_x` and `blocking_y` arguments.

4 Simulations

To assess the proposed methods and compare them with the binary algorithm introduced by Lee et al. (2022), as well as with other existing approaches, we conduct two simulation studies. In the first study, described in Section 4.1, we consider relatively small data sets that can be linked using standard MEC. The second simulation, presented in Section 4.2, uses larger files and aims to test the method combining MEC with the blocking procedure implemented in the `blocking` package.

4.1 Linking Data Sets Using Standard MEC

To compare our continuous approaches with the binary method, we conduct a simulation analogous to Lee et al. (2022, Section 6, *Scenario-II (Informative)*). In this context, *informative* means that the distribution of key variable values depends on whether or not a record should be matched. We choose *Scenario-II* because it is more challenging for record linkage algorithms than *Scenario-I*, which is noninformative. Furthermore, unlike the original simulation, we do not employ the Soundex phonetic algorithm (cf. Knuth, 1973). This is because Soundex is designed specifically for the English language, making it unsuitable as a general-purpose method for record linkage. For other phonetic algorithms, see Howard (2020).

The simulation is based on the `CENSUS`, `CIS`, and `PRD` data sets from McLeod et al. (2011). The variables considered are: `PERNAME1` (first name), `PERNAME2` (last name), `SEX` (gender), `DOB_DAY` (day of birth), `DOB_MON` (month of birth), and `DOB_YEAR` (year of birth). The name variables (`PERNAME1` and `PERNAME2`) are natural candidates for using continuous comparisons. Therefore, we can compare different methods (binary, continuous parametric, and continuous nonparametric) on these two variables. To calculate continuous comparison scores, we use $1 -$ Jaro-Winkler similarity. For the remaining four variables, we consistently

apply the binary approach. For the `SEX` variable, binary comparison is the most natural choice. Regarding the date variables, one could also consider other options (see Vo et al. (2023) for an example of using the Euclidean distance). However, since we do not know the source of errors in these key variables, binary comparisons seem to be the best choice.

A detailed description of the record sampling procedure can be found in Appendix E. We consider three cases, which differ in the true number of matches (n_M) between the two files, set to 400, 250, and 150, respectively. In each case, A and B contain 500 and 1,000 records, respectively.

To compare the methods, we use $\hat{n}_M$, as well as FLR and MMR, given in (1) and (2), respectively. To calculate both error rates, we need information on the true matches, which can be obtained from unique identifiers present in the considered data sets. However, according to Lee et al. (2022), one can also estimate FLR and MMR as follows:

$$\widehat{\text{FLR}} = \frac{1}{\hat{n}_M} \sum_{(a,b)\in\widehat{M}} (1 - \hat{g}_{ab}), \quad \widehat{\text{MMR}} = 1 - \frac{1}{\hat{n}_M} \sum_{(a,b)\in\widehat{M}} \hat{g}_{ab}.$$

Note that, because we treat $\hat{n}_M$ as a plug-in value for n_M, we have $\widehat{\text{FLR}} = \widehat{\text{MMR}}$ by definition.

The simulation results are presented in Table 1 (the names of the methods come from the comparison and estimation approaches applied to the name variables). Notably, the relative performance of the methods is consistent across all three cases. First, the continuous nonparametric approach yielded the best estimates of the number of matches. Nonetheless, it did not result in the lowest error rates. In fact, the highest linkage quality was provided by the continuous parametric method. It yielded the lowest values for both FLR and MMR. Note that in each case, the FLR of the continuous parametric method was below 1%. Furthermore, this method achieved the lowest standard error (SE) of $\hat{n}_M$ in all three cases considered, and its root mean square error (RMSE) was lower than that of the continuous nonparametric approach in the case when $n_M = 150$.

On the other hand, the binary approach resulted in the most biased $\hat{n}_M$. As expected, it led to some information loss, thereby missing the largest fraction of true matches. Although the standard errors of $\hat{n}_M$ for the binary method are lower than those obtained by the continuous nonparametric algorithm, the latter has a clear advantage in terms of RMSE. However, the binary approach performed better than the continuous nonparametric method in terms of FLR. In the first two cases, the differences between the FLR of these approaches were large, but in the third case, both methods yielded similar precision.

Regarding $\widehat{\text{FLR}}$ and $\widehat{\text{MMR}}$, the results demonstrate that these estimators are not very accurate. They preserve the relative changes between the cases, but we do not recommend treating these estimators as reliable. This is particularly evident because, as shown in our simulation, the true FLR and MMR can diverge substantially from each other, as can $\hat{n}_M$ and n_M.

Note that, in general, a lower n_M results in higher error rates due to the increased difficulty of the task, since Ω is even more dominated by nonmatches. Regardless of n_M, in our simulation, the continuous parametric method resulted in the highest linkage quality. This shows that continuous comparisons can substantially improve the linkage of data sets containing small discrepancies.

n_M	MEC Method	$\hat{n}_M$ (SE)	RMSE($\hat{n}_M$)	FLR	MMR	$\widehat{\text{FLR}}$	$\widehat{\text{MMR}}$
400	Binary	382.2 (6.5)	19.0	0.62	5.05	3.93	3.93
	C. param.	385.5 (6.1)	15.7	0.21	3.84	1.09	1.09
	C. nonparam.	388.4 (8.3)	14.2	1.21	4.08	3.59	3.59
250	Binary	237.6 (5.2)	13.5	1.52	6.43	4.39	4.39
	C. param.	240.2 (4.0)	10.6	0.56	4.51	1.16	1.16
	C. nonparam.	243.7 (6.4)	9.0	2.36	4.84	3.89	3.89
150	Binary	141.1 (4.8)	10.1	2.85	8.66	5.65	5.65
	C. param.	143.7 (3.1)	7.0	0.96	5.17	1.21	1.21
	C. nonparam.	144.1 (5.3)	7.9	2.88	6.74	4.21	4.21

Table 1: True number of matches, average estimates with their standard errors (SE) and root mean square errors (RMSE), and average linkage error rates along with their estimates across 100 simulations. The error rates are presented as percentage values. The names of the MEC methods come from the comparison and estimation approaches applied to the name variables.

4.2 Linking Data Sets Using Blocked MEC

In this simulation, we demonstrate MEC record linkage after applying blocking. To this end, we use 100 pairs of data sets. These are based on a fictional foreign-born population in Poland, generated using publicly available information and preserving the distributions of administrative registers. In each pair, A and B contain 9,000 and 8,000 records, respectively, and the number of matches is equal to 7,000. Note that considering the full comparison space Ω would require 7.2×10^7 comparisons in each iteration.

The variables considered are: `FNAME` (first name), `SURNAME` (surname), `DOB_DAY` (day of birth), `DOB_MON` (month of birth), `DOB_YEAR` (year of birth), and `COUNTRY` (country). The first five are used both for blocking and final linkage, while the last one serves only for blocking. Additionally, information on true matches can be extracted using the unique identifier `TRUE_ID`.

Here, we use the Nearest Neighbor Descent Method (NND; Dong et al., 2011), implemented in the `rnndescent` R package (Melville, 2026b) and available via the `blocking` package. The same blocking configuration is used for all record linkage methods compared. To provide input for the blocking algorithm, we concatenate information from all six key variables considered and use spaces between the values. Then, we represent the input data as 3-character shingles (sequences). For example, `"Thomas"` would be represented as `("Tho", "hom", "oma", "mas")`. Using 3-character shingles instead of the default 2-character ones yields a larger number of blocked pairs. This reduces the MMR at the blocking stage. For the same purpose, we set the `epsilon` parameter of the NND method to 0.5. See Beręsewicz and Struzik (2026) for an example of modifying this parameter value.

As in the simulation in Section 4.1, in the MEC stage, we apply binary comparisons to the `DOB` variables. Analogously, `FNAME` and `SURNAME` are used to evaluate the continuous

Method	$\hat{n}_M$	FLR (%)	MMR (%)
MEC (binary, $\rho = 0$)	7,218.8	3.19	0.16
MEC (binary, $\rho = 0.5$)	7,218.9	3.43	0.41
MEC (mixed, $\rho = 0$)	6,744.1	0.01	3.66
MEC (mixed, $\rho = 0.5$)	6,882.5	0.14	1.81
FS	6,903.5	1.59	2.95

Table 2: Average estimates of the number of matches and average error rates across 100 simulations. The true number of matches is 7,000.

approach against the binary algorithm. However, in this simulation, we apply only the continuous parametric method, as it performed better in the previous simulation and is substantially faster than the nonparametric algorithm. Here, we denote the approach based on continuous comparisons and the parametric method for the name variables by *mixed*. Finally, we can vary the value of ρ. We consider $\rho = 0$ (which is equivalent to using all pairs from $U_B^{(t)}$ to estimate $\boldsymbol{\xi}$ in each iteration) and $\rho = 0.5$. In total, this results in four MEC-based methods.

We also use the method by Fellegi and Sunter (FS; 1969) and its implementation via the `reclin2` R package (van der Laan, 2022, 2025). To automatically match pairs using the FS method, we perform a greedy selection of pairs whose model-assigned weights are greater than 0. Note that the FS model uses agreement/disagreement comparisons. However, one can also construct a binary indicator based on a continuous measure. For instance, one can calculate the Jaro-Winkler similarity score (JW) and set a threshold above which a pair of values is treated as an agreement. See Appendix F for results obtained using this approach.

In total, we compare five methods, and the mean values of $\hat{n}_M$, FLR, and MMR are presented in Table 2. In terms of estimating the number of matches, the estimates closest to the true number of matches (although still underestimated) were obtained by the FS model. The mixed MEC approach tended to underestimate n_M even more. However, changing the value of ρ from 0 to 0.5 substantially improved $\hat{n}_M$ (from 6,744.1 to 6,882.5). In contrast, the binary method overestimated the number of matches and was almost unaffected by the modification of ρ.

Regarding the FLR, the lowest values were obtained by the mixed MEC approach. Notably, with $\rho = 0$, it yielded an FLR equal to 0.01%. Even after the change in ρ, the FLR was still very low, that is, 0.14%. Meanwhile, the FS model resulted in an FLR of 1.59%, and the binary MEC falsely predicted more than 3% of matches in both cases. These results show that the continuous parametric approach can help to achieve an FLR very close to zero, which is not the case with any other method considered.

However, one should also take the MMR into account. Here, the lowest values were yielded by the binary MEC. The mixed variant initially missed 3.66% of the true matched pairs, but after the modification of ρ, this error rate decreased substantially to 1.81%, thereby outperforming the FS model, which yielded an MMR of 2.95%.

One can draw several conclusions from the simulation. If the aim of linking records is to minimize the FLR while still finding a reasonable number of matches (which is the case in most applications), the mixed method might be a better choice than the other approaches considered. Additionally, one can manipulate the value of ρ, which (as demonstrated in our simulation) can substantially decrease the MMR of the mixed MEC while preserving a low FLR. On the other hand, in the FS model, the linkage error rates are closer to each other, but neither of them is close to 0.

Regarding the binary MEC approach, it might be useful in contexts where one wants to minimize the MMR. However, this is rarely the desired outcome, as we do not want to incorporate false information into a subsequent analysis. Additionally, it should be mentioned that the change in ρ increased both error rates obtained by this method. Therefore, when using the binary approach, applying the default value of $\rho = 0$ might be a reasonable choice.

In each iteration, blocking yielded a comparison space consisting of 8,000 pairs (which is equal to $\min\{|A|, |B|\}$). Consequently, the computational burden was reduced by a factor of 9,000 compared to evaluating all possible record pairs. This is a substantial improvement, indicating the high potential of combining MEC with blocking.

5 Application to Polish Register Data

In this section, we apply the proposed method to Polish register data on foreign-born nationals for 2023. We use two registers held by the Office for Foreigners (UDSC) and the Social Insurance Institution (ZUS). Poland has a population register, PESEL, but not all migrants apply for a PESEL identifier, and there are numerous records in these registers that do not contain identifiers.

The first register (UDSC) consists of all documents that were issued to foreign-born nationals who applied for legalization of stay; for more details, see Office for Foreigners (2026). Note that this register covers all valid documents regardless of the actual presence in Poland and may contain duplicates, as one person could have more than one document. The second register (ZUS) refers to all foreigners who were registered by their employers for pension and disability insurance; for more details, see the definitions in Zakład Ubezpieczeń Społecznych (2026, in Polish). The ZUS register covers a substantial part of the labor market in Poland, in particular: employment contracts, contracts of mandate, and registered unemployment. Self-employment and contracts for specific work (Pol. *umowa o dzieło*) are not covered in this register.

In this study, we focus on the population aged 15–89, which is consistent with the International Labour Organization definition (used, for instance, in the Labour Force Survey). Thus, linking these two data sources allows us to estimate labor market activity indicators related to foreign-born nationals who applied for legalization of stay. Table 3 presents basic information about these registers before and after internal deduplication. About 83% and 90% of individuals from the UDSC and ZUS registers, respectively, do not have identifiers. The overlap between the two sources, identified by the PESEL identifier, amounts to 322,929 individuals, including 155,830 who applied for a legal stay in 2023.

Given the availability of the PESEL identifiers, we are able to assess the quality of the discussed methods. In particular, we consider the FS model with binary comparisons, MEC

Step	Description	UDSC	ZUS
1	Initial (before internal deduplication)	701,830	1,206,357
2	After deduplication (PESEL, unique name and birth date)	700,990	1,106,080
3	15–89 population	650,084	1,104,303
4	15–89 population and applied in 2023	268,681	—
5	Number of persons with PESEL identifiers in Step 3	536,808	991,359

Table 3: Basic information about individuals covered by the registers used in the study (as of December 31, 2023).

with binary comparisons (denoted by MEC-B), and MEC with mixed comparisons (MEC-M). Binary comparisons in all methods are applied to the month and day of birth, as well as the country code. The treatment of first and last names, as well as their concatenation, varies across the approaches. Specifically, in the FS and MEC-B models, we use binary comparisons, while in MEC-M, we apply continuous comparisons.

Due to the large number of comparisons, we first block records by sex and birth year (creating a separate group for people born in 1950 or earlier, which results in 118 blocks in total). Then, within each block, we use the HNSW algorithm (Malkov and Yashunin, 2020), implemented in the `RcppHNSW` package (Melville, 2026a) and available via the `blocking` package, to further reduce the comparison space. The blocking variable is created by concatenating the first and last names (cleared of all diacritic marks, spaces, or other non-alphabetic characters), sex, date of birth, and country code (ISO 3166). Based on the PESEL identifiers, we identify that this approach misses 0.06% of true matches. However, the exact error rate for the non-PESEL population remains unknown.

In the main text, we report the results obtained based on exact binary comparisons, whereas Appendix G presents the results based on binary comparisons with a JW threshold set to 0.95. Table 4 contains the results of the linkage procedure of the UDSC and ZUS registers for all individuals and applicants for a legal stay in 2023. All three methods performed well on the PESEL-verifiable subset, but they were characterized by different balances between false and missed matches. MEC-B linked the most records (411,878) and missed the fewest true matches (an MMR of 0.84%), but it also had the highest FLR (0.15%). MEC-M was the most conservative: its FLR was the lowest (0.06%), but it missed 3.88% of true matches. FS fell in between. Because the FLR differed by less than 0.1 percentage points across the methods, while the MMR varied more than fourfold, the choice of method matters mainly for how many matches are lost. The pattern was the same among applicants for a legal stay in 2023, with a slightly lower MMR for every method. Note that all rates are computed only for the records with a PESEL in both registers, providing limited insight into the links found beyond that subset.

We investigate whether the FLR for the FS and MEC methods may actually be higher than reported in Table 4, as suggested by the simulations in Section 4. For this purpose, we assume that records with identical names and dates of birth refer to the same person, and compare this criterion with PESEL agreement. The results are presented in Table 5.

Measure	FS	MEC-B	MEC-M
All persons aged 15–89			
Linked records	362,379	411,878	372,779
Linked records with PESEL in both registers	312,527	319,705	310,227
FLR, PESEL-based	0.07%	0.15%	0.06%
MMR, PESEL-based	3.22%	0.84%	3.88%
Applicants for a legal stay in 2023			
Linked records	170,297	189,015	173,970
Linked records with PESEL in both registers	151,316	154,451	149,998
FLR, PESEL-based	0.07%	0.14%	0.06%
MMR, PESEL-based	2.90%	0.73%	3.68%

Table 4: Linkage of the UDSC and ZUS registers using exact (binary) comparisons: foreign-born nationals aged 15–89, 2023.

First name, last name, and birth date	FS	MEC-B	MEC-M
Conflicting PESEL identifiers			
Identical	71	55	55
Not identical	160	430	120
Missing PESEL in at least one register			
Identical	44,695	45,047	44,784
Not identical	12,501	38,932	11,488

Table 5: Linked pairs using exact (binary) comparisons by PESEL match status and agreement on first name, last name, and birth date, population aged 15–89.

Among the linked pairs with conflicting PESEL identifiers, only a few share names and dates of birth (71 for FS, 55 for MEC-B and MEC-M); thus, the reported FLR for the PESEL-based subpopulation is not understated. The greater concern pertains to pairs without PESEL, which cannot be verified against the reference identifier. Approximately 20–22% of the FS and MEC-M links differ in the first name, last name, or date of birth, compared to 46% for MEC-B. Although some of these differences may reflect data entry errors or name changes rather than false links, the substantial gap suggests that the higher recall of MEC-B comes at a cost in precision that the PESEL-based FLR does not capture.

6 Final Remarks

We have generalized the unsupervised algorithm for record linkage, introduced by Lee et al. (2022). Our approach allows one to compare selected key variables using continuous functions, such as string similarity measures. We have proposed two methods for handling continuous comparisons: parametric and nonparametric. The former is based on the hurdle Gamma distribution, while the latter uses the Kullback-Leibler Importance Estimation Procedure. In addition, we have shown how to combine the idea of maximum entropy record linkage with probabilistic blocking of records that yields numerous small blocks. Finally, we have implemented all the approaches considered in the `automatedRecLin` R package, available on CRAN.

Our simulations have shown that incorporating continuous comparisons can substantially improve linkage results. In particular, when linking data sets based on the full comparison space, the continuous parametric method applied to names outperforms the binary algorithm in terms of the error rates and the estimation of the number of matches. Additionally, the continuous nonparametric approach performs worse than the binary method only regarding FLR.

When MEC is applied after probabilistic blocking, a tradeoff can be observed. Using continuous comparisons minimizes the FLR, whereas agreement/disagreement indicators minimize the MMR. Meanwhile, the FS model yields balanced error rates, but neither is close to 0.

We have applied the proposed method to data from two administrative registers in Poland to study the activity of migrants in the Polish labor market. Although we have not been able to fully assess the results due to missing unique identifiers, our investigation has suggested conclusions similar to those drawn in the simulation studies.

An important future research direction is to further examine the convergence of the unsupervised approach or potentially modify the algorithm to guarantee a finite number of iterations (see Appendix A for some initial work). Moreover, one could consider incorporating continuous distributions different from the hurdle Gamma distribution into the MEC framework. Discrete key variables with multiple possible values and different levels of agreement between them could also be modeled. Finally, to further improve the scalability of the algorithms, the `automatedRecLin` R package could be integrated with DuckDB (an efficient database management system) via an interface provided by the `duckdb` package (Mühleisen and Raasveldt, 2026).

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The paper's topic was presented by Adam Struzik at the following conferences: Student Scientific Conference ElementsX 2025 (Kraków, Poland), useR! Conference 2026 (Warsaw, Poland), and Joint Statistical Meetings 2026 (Boston, Massachusetts, USA).

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Appendix A. Convergence of the Unsupervised Algorithm

The supplementary material of Lee et al. (2022, Appendix A, available at arXiv:2009.14797) provides Theorem 1, which aims to establish the convergence of the binary MEC algorithm in the unsupervised setting. However, the proof contains two errors that leave the statement unproven.

Firstly, the third term after applying Jensen's inequality should have $1/n_M^{(t+1)}$ inside the logarithm rather than outside, leading to:

$$-\left|1-\frac{n_M^{(t+1)}}{n_M^{(t)}}\right|\log\left\{\frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)}\frac{m^{(t)}(\boldsymbol{\gamma}_{ab})}{u^{(t)}(\boldsymbol{\gamma}_{ab})}\right\}.$$

Note that this is likely a typographical error.

However, a critical mathematical problem occurs in the third inequality. Given that $m^{(t+1)}(\boldsymbol{\gamma}), u^{(t)}(\boldsymbol{\gamma}) \in [0,1]$, their inverses are bounded below by 1. It follows that:

$$\frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)}\frac{m^{(t)}(\boldsymbol{\gamma}_{ab})}{m^{(t+1)}(\boldsymbol{\gamma}_{ab})} \geq \frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)} m^{(t)}(\boldsymbol{\gamma}_{ab}),$$

as well as

$$\frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)}\frac{u^{(t+1)}(\boldsymbol{\gamma}_{ab})}{u^{(t)}(\boldsymbol{\gamma}_{ab})} \geq \frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)} u^{(t+1)}(\boldsymbol{\gamma}_{ab})$$

and

$$\frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)}\frac{m^{(t)}(\boldsymbol{\gamma}_{ab})}{u^{(t)}(\boldsymbol{\gamma}_{ab})} \geq \frac{1}{n_M^{(t+1)}}\sum_{(a,b)\in\Omega} g_{ab}^{(t+1)} m^{(t)}(\boldsymbol{\gamma}_{ab}).$$

Because $\log(\cdot)$ is a strictly increasing function, $-\log(\cdot)$ is strictly decreasing. Therefore, applying the negative logarithm to these terms requires inverting the inequality sign in the proof in Lee et al. (2022, Appendix A).

The transition proposed in Lee et al. (2022, Appendix A) works if and only if the expressions are equal, which requires $m^{(t+1)}(\boldsymbol{\gamma}_{ab}) = u^{(t)}(\boldsymbol{\gamma}_{ab}) = 1$ for all $(a,b) \in \Omega$ such that $g_{ab}^{(t+1)} = 1$. This condition is not true in practical record linkage scenarios, making the proof incorrect.

Due to the gap in the proof, it cannot be formally guaranteed whether the binary unsupervised algorithm converges in a finite number of iterations; likewise, this applies to the continuous methods. However, in our extensive simulation studies and real-world application to Polish register data, as well as during the testing of the `automatedRecLin` package, we did not encounter any convergence issues.

Nonetheless, this convergence problem warrants further investigation. On the one hand, one can look for sufficient conditions that would guarantee the convergence of the procedure. Ideally, such conditions would be verifiable before the start of the algorithm, based on either the raw data or (most likely) on the comparison vectors. On the other hand, the method itself could be modified to structurally enforce convergence. For instance, one could introduce dynamic boundaries on $\hat{n}_M$ that tighten whenever the algorithm changes direction, thereby preventing it from entering an infinite loop.

Appendix B. Detailed Continuous Parametric Algorithm

Assume that the components of $\boldsymbol{\gamma}$ are conditionally independent (both in M and U). Let $\gamma^k \mid M$ and $\gamma^k \mid U$ follow hurdle Gamma distributions with parameters $p_{0M}^k, \alpha_M^k, \beta_M^k$ and $p_{0U}^k, \alpha_U^k, \beta_U^k$, respectively. Let $\boldsymbol{\theta}$ and $\boldsymbol{\xi}$ denote the parameter vectors associated with M and U, respectively. In this setting, (4) holds.

Assume a common support $\mathcal{S}(M) = \mathcal{S}(U)$. Let M_1 denote the maximal MEC set. The continuous parametric MEC algorithm for record linkage is detailed below.

1. Set initial parameters

$$\boldsymbol{\theta}^{(0)} = (p_{0M}^{1(0)}, \alpha_M^{1(0)}, \beta_M^{1(0)}, p_{0M}^{2(0)}, \alpha_M^{2(0)}, \beta_M^{2(0)}, \ldots, p_{0M}^{K(0)}, \alpha_M^{K(0)}, \beta_M^{K(0)})$$

   and set $M^{(0)} = M_1$.

2. Calculate

$$\hat{p}_{0U}^k = \frac{1}{n} \sum_{(a,b)\in\Omega} \mathbb{I}(\gamma_{ab}^k = 0).$$

   Denote

$$\bar{\gamma}_+^k = \frac{\sum_{(a,b)\in\Omega} \mathbb{I}(\gamma_{ab}^k > 0)\gamma_{ab}^k}{\sum_{(a,b)\in\Omega} \mathbb{I}(\gamma_{ab}^k > 0)}.$$

   $\hat{\alpha}_U^k$ is given by the solution (found using iterative methods) of the following equation:

$$\sum_{(a,b)\in\Omega} \mathbb{I}(\gamma_{ab}^k > 0)\Big[\log \gamma_{ab}^k - \log \bar{\gamma}_+^k - \psi(\alpha_U^k) + \log \alpha_U^k\Big] = 0,$$

   where

$$\psi(z) = \frac{\partial}{\partial z} \log \Gamma(z)$$

   is the digamma function. Calculate

$$\hat{\beta}_U^k = \frac{\hat{\alpha}_U^k}{\bar{\gamma}_+^k}.$$

3. Given

$$\hat{\boldsymbol{\xi}} = (\hat{p}_{0U}^1, \hat{\alpha}_U^1, \hat{\beta}_U^1, \hat{p}_{0U}^2, \hat{\alpha}_U^2, \hat{\beta}_U^2, \ldots, \hat{p}_{0U}^K, \hat{\alpha}_U^K, \hat{\beta}_U^K),$$

   calculate

$$r^{(0)}(\boldsymbol{\gamma}_{ab}) = \frac{m(\boldsymbol{\gamma}_{ab}; \boldsymbol{\theta}^{(0)})}{u(\boldsymbol{\gamma}_{ab}; \hat{\boldsymbol{\xi}})}$$

   and

$$n_M^{(0)} = \sum_{(a,b)\in\Omega} \hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}),$$

   where

$$\hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}) = \min\Big\{\frac{|M^{(0)}| r^{(0)}(\boldsymbol{\gamma}_{ab})}{|M^{(0)}|(r^{(0)}(\boldsymbol{\gamma}_{ab}) - 1) + n}, 1\Big\}.$$

4. Find the MEC set $M^{(1)}$ such that $|M^{(1)}| = n_M^{(0)}$ by deduplication in descending order of $r^{(0)}(\boldsymbol{\gamma}_{ab})$ over Ω.

5. For the t-th iteration ($t \geq 1$), let $g_{ab}^{(t)} = \mathbb{I}((a,b) \in M^{(t)})$.

   (a) Update
   $$\hat{p}_{0M}^{k(t)} = \frac{1}{|M^{(t)}|} \sum_{(a,b)\in\Omega} g_{ab}^{(t)} \mathbb{I}(\gamma_{ab}^k = 0).$$
   Calculate
   $$\bar{\gamma}_{+}^{k(t)} = \frac{\sum_{(a,b)\in\Omega} g_{ab}^{(t)} \mathbb{I}(\gamma_{ab}^k > 0)\gamma_{ab}^k}{\sum_{(a,b)\in\Omega} g_{ab}^{(t)} \mathbb{I}(\gamma_{ab}^k > 0)}$$
   and obtain $\hat{\alpha}_M^{k(t)}$ as the solution of the following equation:
   $$\sum_{(a,b)\in\Omega} g_{ab}^{(t)} \mathbb{I}(\gamma_{ab}^k > 0)\Big[\log \gamma_{ab}^k - \log \bar{\gamma}_{+}^{k(t)} - \psi(\alpha_M^k) + \log \alpha_M^k\Big] = 0.$$
   Update
   $$\hat{\beta}_M^{k(t)} = \frac{\hat{\alpha}_M^{k(t)}}{\bar{\gamma}_{+}^{k(t)}}.$$

   (b) Calculate
   $$n_M^{(t)} = \sum_{(a,b)\in\Omega} \hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}),$$
   where
   $$\hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}) = \min\Big\{\frac{|M^{(t)}| r^{(t)}(\boldsymbol{\gamma}_{ab})}{|M^{(t)}|(r^{(t)}(\boldsymbol{\gamma}_{ab}) - 1) + n}, 1\Big\}$$
   and
   $$r^{(t)}(\boldsymbol{\gamma}_{ab}) = \frac{m(\boldsymbol{\gamma}_{ab}; \boldsymbol{\theta}^{(t)})}{u(\boldsymbol{\gamma}_{ab}; \hat{\boldsymbol{\xi}})}.$$

   (c) Find the MEC set $M^{(t+1)}$ such that $|M^{(t+1)}| = n_M^{(t)}$ by deduplication in descending order of $r^{(t)}(\boldsymbol{\gamma}_{ab})$ over Ω.

6. Repeat Step 5 until $|n_M^{(t)} - n_M^{(t-1)}| < \delta$ or $\|\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^{(t-1)}\| < \varepsilon$, where δ and ε are pre-defined thresholds (small positive values).

Appendix C. Detailed Continuous Nonparametric Algorithm

Let the density ratio r be given by (5). Assume a common support $\mathcal{S}(M) = \mathcal{S}(U)$. The continuous nonparametric MEC algorithm for record linkage is detailed below.

1. Set initial density ratio $r^{(0)}(\boldsymbol{\gamma}_{ab})$ and set $M^{(0)} = M_1$.

2. Calculate
$$n_M^{(0)} = \sum_{(a,b)\in\Omega} \hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}),$$
where
$$\hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}) = \min\Big\{\frac{|M^{(0)}|r^{(0)}(\boldsymbol{\gamma}_{ab})}{|M^{(0)}|(r^{(0)}(\boldsymbol{\gamma}_{ab})-1)+n}, 1\Big\}.$$

3. Find the MEC set $M^{(1)}$ such that $|M^{(1)}| = n_M^{(0)}$ by deduplication in descending order of $r^{(0)}(\boldsymbol{\gamma}_{ab})$ over Ω.

4. For the t-th iteration ($t \geq 1$), let $g_{ab}^{(t)} = \mathbb{I}((a,b) \in M^{(t)})$.

   (a) Calculate
   $$\hat{p}_{0M}^{k(t)} = \frac{1}{|M^{(t)}|} \sum_{(a,b)\in\Omega} g_{ab}^{(t)} \mathbb{I}(\gamma_{ab}^k = 0)$$
   and
   $$\hat{p}_{0U}^{k(t)} = \frac{1}{|\Omega \setminus M^{(t)}|} \sum_{(a,b)\in\Omega} (1 - g_{ab}^{(t)}) \mathbb{I}(\gamma_{ab}^k = 0).$$
   For the samples $\{\gamma_{ab}^k \colon (a,b) \in M^{(t)}, \gamma_{ab}^k > 0\}$ and $\{\gamma_{ab}^k \colon (a,b) \in \Omega \setminus M^{(t)}, \gamma_{ab}^k > 0\}$, estimate the density ratio $r_+^{k(t)}(\cdot)$ using the KLIEP method.

   (b) Update
   $$r^{(t)}(\boldsymbol{\gamma}_{ab}) = \prod_{k=1}^{K} \left\{ \left(\frac{\hat{p}_{0M}^{k(t)}}{\hat{p}_{0U}^{k(t)}} \right)^{\mathbb{I}(\gamma_{ab}^k = 0)} \left(\frac{1 - \hat{p}_{0M}^{k(t)}}{1 - \hat{p}_{0U}^{k(t)}} r_+^{k(t)}(\gamma_{ab}^k) \right)^{\mathbb{I}(\gamma_{ab}^k > 0)} \right\}.$$
   Calculate
   $$n_M^{(t)} = \sum_{(a,b)\in\Omega} \hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}),$$
   where
   $$\hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}) = \min\Big\{\frac{|M^{(t)}|r^{(t)}(\boldsymbol{\gamma}_{ab})}{|M^{(t)}|(r^{(t)}(\boldsymbol{\gamma}_{ab})-1)+n}, 1\Big\}.$$

   (c) Find the MEC set $M^{(t+1)}$ such that $|M^{(t+1)}| = n_M^{(t)}$ by deduplication in descending order of $r^{(t)}(\boldsymbol{\gamma}_{ab})$ over Ω.

5. Repeat Step 4 until $|n_M^{(t)} - n_M^{(t-1)}| < \delta$, where δ is a pre-defined threshold (a small positive value).

Appendix D. Detailed Blocked MEC for Record Linkage

Suppose that a blocking procedure returns H blocks $\Omega_1, \Omega_2, \ldots, \Omega_H$, as discussed in Section 3.3.1. Assume that $\gamma^k \mid M_B$ and $\gamma^k \mid U_B$ follow Bernoulli or hurdle Gamma distributions. Let $\boldsymbol{\theta}$ and $\boldsymbol{\xi}$ denote the parameter vectors associated with M_B and U_B, respectively.

Assume a common support $\mathcal{S}(M_B) = \mathcal{S}(U_B)$. Set a value $\rho \in [0, 1)$. The blocked unsupervised MEC algorithm for record linkage is detailed below.

1. Initialize $M_B^{(0)}$ as the first set of size ν obtained by deduplication in ascending order of d_{ab} over Ω_B. Calculate $U_B^{(0)} = \Omega_B \setminus M_B^{(0)}$.

2. Estimate $\boldsymbol{\theta}$ and $\boldsymbol{\xi}$ by using $M_B^{(0)}$ and $U_B^{(0)}$, respectively. Denote the estimates as $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\xi}^{(0)}$, respectively.

3. Given $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\xi}^{(0)}$, calculate

$$s^{(0)}(\boldsymbol{\gamma}_{ab}) = \frac{u(\boldsymbol{\gamma}_{ab}; \boldsymbol{\xi}^{(0)})}{m(\boldsymbol{\gamma}_{ab}; \boldsymbol{\theta}^{(0)})}$$

and

$$n_{M_B}^{(0)} = \sum_{(a,b)\in\Omega_B} \hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}),$$

where

$$\hat{g}^{(0)}(\boldsymbol{\gamma}_{ab}) = 1 - \min\Big\{ \frac{|U_B^{(0)}| s^{(0)}(\boldsymbol{\gamma}_{ab})}{|U_B^{(0)}|(s^{(0)}(\boldsymbol{\gamma}_{ab}) - 1) + N_B}, 1 \Big\}.$$

4. Find the MEC set $M_B^{(1)}$ such that $|M_B^{(1)}| = n_{M_B}^{(0)}$ by deduplication in ascending order of $s^{(0)}(\boldsymbol{\gamma}_{ab})$ over Ω_B.

5. For the t-th iteration ($t \geq 1$), let $g_{ab}^{(t)} = \mathbb{I}((a,b) \in M_B^{(t)})$.

   (a) Update $\boldsymbol{\theta}^{(t)}$ using $M_B^{(t)}$. Take a subset $U_B'^{(t)} \subseteq U_B^{(t)} = \Omega_B \setminus M_B^{(t)}$ consisting of $\lfloor \rho |U_B^{(t)}| \rfloor$ pairs with the highest values of $\hat{g}_{ab}^{(t-1)}$ and use $U_B^{(t)} \setminus U_B'^{(t)}$ to compute $\boldsymbol{\xi}^{(t)}$.

   (b) Calculate

$$n_{M_B}^{(t)} = \sum_{(a,b)\in\Omega_B} \hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}),$$

   where

$$\hat{g}^{(t)}(\boldsymbol{\gamma}_{ab}) = 1 - \min\Big\{ \frac{|U_B^{(t)}| s^{(t)}(\boldsymbol{\gamma}_{ab})}{|U_B^{(t)}|(s^{(t)}(\boldsymbol{\gamma}_{ab}) - 1) + N_B}, 1 \Big\}$$

   and

$$s^{(t)}(\boldsymbol{\gamma}_{ab}) = \frac{u(\boldsymbol{\gamma}_{ab}; \boldsymbol{\xi}^{(t)})}{m(\boldsymbol{\gamma}_{ab}; \boldsymbol{\theta}^{(t)})}.$$

   (c) Find the MEC set $M_B^{(t+1)}$ such that $|M_B^{(t+1)}| = n_{M_B}^{(t)}$ by deduplication in ascending order of $s^{(t)}(\boldsymbol{\gamma}_{ab})$ over Ω_B.

6. Repeat Step 5 until $|n_{M_B}^{(t)} - n_{M_B}^{(t-1)}| < \delta$ or $\|(\boldsymbol{\theta}^{(t)}, \boldsymbol{\xi}^{(t)}) - (\boldsymbol{\theta}^{(t-1)}, \boldsymbol{\xi}^{(t-1)})\| < \varepsilon$, where δ and ε are pre-defined thresholds (small positive values).

Data set	Description
`CENSUS` (25,343)	A fictional data set to represent some observations from a Decennial Census
`CIS` (24,613)	Fictional observations from Customer Information System, combined administrative data from the tax and benefit systems
`PRD` (24,750)	Fictional observations from Patient Register Data of the National Health Service

Table 6: Description of the data sets used in Section 4.1 (size in parentheses).

Appendix E. Sampling Details

The description of the record sampling procedure from Section 4.1 is based on Lee et al. (2022, Section 6, *Scenario-II (Informative)*). The simulation uses the `CENSUS`, `CIS`, and `PRD` data sets from McLeod et al. (2011), which are outlined in Table 6.

For sampling, we use the unique identification variable (`PERSON_ID`), which is available in all three data sets. We sample $n_A = 500$ and $n_B = 1{,}000$ records from `PRD` and `CIS`, respectively. Let p_A be the proportion of records in the smaller file that are also selected in the larger file. This parameter allows us to vary the degree of overlap, that is, the set of matched individuals. We consider $p_A \in \{0.8, 0.5, 0.3\}$. The sampling procedure is as follows.

1. Sample n_A entities from $\texttt{CENSUS} \cap \texttt{PRD} \cap \texttt{CIS}$ and denote this set by A.
2. Sample $n_M = n_A p_A$ entities from A to serve as the matched individuals, and denote this subset by AB.
3. Sample $n_B - n_M$ entities from $\texttt{CIS} \setminus A$ having $\texttt{SEX} = \text{F}$, $\texttt{YEAR} \leq 1970$, and odd `DOB_MON`. Denote this set by B_0, and let $B = AB \cup B_0$.
4. Represent A and B by the associated records from `PRD` and `CIS`, respectively.

Appendix F. Extended Results of the Simulation with Blocking

We present extended results of the simulation study in Section 4.2. As mentioned before, we can construct a binary comparator based on a continuous measure and a threshold. The comparison yields a 1 if the continuous value is greater than the threshold and a 0 otherwise. Such comparisons are suitable for names, and we treat them as input to the binary MEC algorithm and the FS model. Because we can modify the value of ρ in MEC record linkage, we now have 3 new combinations of approaches, which we label as *with JW threshold*. In all cases, we set the threshold to 0.95.

The results are shown in Table 7. In terms of the binary MEC, modifying the comparisons barely changed the results. However, the newly applied FS model yielded the least biased mean value of $\hat{n}_M$ among all approaches considered. The error rates became relatively balanced, but neither is strictly minimized.

Method	$\hat{n}_M$	FLR (%)	MMR (%)
MEC (binary, $\rho = 0$)	7,218.8	3.19	0.16
MEC (binary, $\rho = 0.5$)	7,218.9	3.43	0.41
MEC (binary, with JW threshold, $\rho = 0$)	7,218.6	3.18	0.15
MEC (binary, with JW threshold, $\rho = 0.5$)	7,218.9	3.43	0.41
MEC (mixed, $\rho = 0$)	6,744.1	0.01	3.66
MEC (mixed, $\rho = 0.5$)	6,882.5	0.14	1.81
FS	6,903.5	1.59	2.95
FS (with JW threshold)	7,010.9	1.83	1.67

Table 7: Average estimates of the number of matches and average error rates across 100 simulations. The true number of matches is 7,000.

The results suggest that constructing a binary indicator based on the Jaro-Winkler similarity measure can substantially improve the performance of the FS model, which is not the case with the binary MEC approach. If the goal is not to strictly minimize a specific error rate, the FS model might be a reasonable approach. Otherwise, one of the MEC algorithms can be a better option.

Appendix G. Extended Results of the Case Study

Measure	FS	MEC-B	MEC-M
All persons aged 15–89			
Linked records	366,172	411,778	372,794
Linked records with PESEL in both registers	316,320	317,478	310,227
FLR, PESEL-based	0.08%	0.17%	0.06%
MMR, PESEL-based	2.05%	1.52%	3.88%
Applicants for a legal stay in 2023			
Linked records	171,960	188,883	173,974
Linked records with PESEL in both registers	152,979	153,380	149,998
FLR, PESEL-based	0.08%	0.18%	0.06%
MMR, PESEL-based	1.83%	1.40%	3.68%

Table 8: Linkage of the UDSC and ZUS registers using Jaro-Winkler comparisons (with a threshold of 0.95): foreign-born nationals aged 15–89, 2023.

First name, last name, and birth date	FS	MEC-B	MEC-M
Conflicting PESEL identifiers			
Identical	106	54	55
Not identical	130	479	120
Missing PESEL in at least one register			
Identical	44,691	43,085	44,784
Not identical	13,388	33,916	11,488

Table 9: Linked pairs using Jaro-Winkler comparisons (with a threshold of 0.95) by PESEL match status and agreement on first name, last name, and birth date, population aged 15–89.

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